Write $P = (X, f, (a, b), E)$ for the ambient space, continuous map, endpoints, and endpoint selection. The selection has four values: none, lower, upper, both. In the following notation, regard E as its selected subset of $\{a, b\}$. The expanded pair $X_r(P)$ has components $f^{-1}[a - r, b + r]$ and $f^{-1}(\bigcup_{e \in E}[e - r, e + r])$. The shifted pair $T_t(P)$ has components $f^{-1}[a + t, b + t]$ and $f^{-1}(E + t)$. For a perturbation h, $A(P, h)$ has components $h^{-1}[a, b]$ and $h^{-1}(E)$. Pair inclusion means inclusion of both components. These results establish set inclusions and certified pair inclusions; they do not assert the well-group characterization. A certified pair Q represents pair data S when their ambient subsets agree and their relative subsets agree. A certified inclusion stores the nesting and both component-inclusion proofs.

Result 1 (endpoint choices are exhaustive). [Topology foundations, two Boolean endpoint flags]
Let E be a selection of the lower and upper endpoints. Then

$$E = \text{none or } E = \text{lower or } E = \text{upper or } E = \text{both.}$$

Result 2 (selected endpoints lie in the interval). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let I be a pair of real endpoints. Let E be a selection of the lower and upper endpoints. Assume $I.\text{lower} \leq I.\text{upper}$ (`order`). Then

$$E(I, E) \subseteq [I.lower, I.upper].$$

Adapted from the cited source. Scope: endpoint inclusion.

Result 3 (expanded pair is nested). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let P be interlevel problem data. Let r be a real number. Assume $P.\text{interval.lower} \leq P.\text{interval.upper}$ (`order`). Assume $r \geq 0$ (`radius`). Then

$$X_r(P) \text{ is a nested pair.}$$

Adapted from the cited source. Scope: pair inclusion.

Result 4 (shifted pair is nested). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let P be interlevel problem data. Let t be a real number. Assume $P.\text{interval.lower} \leq P.\text{interval.upper}$ (`order`). Then

$$T_t(P) \text{ is a nested pair.}$$

Adapted from the cited source. Scope: pair inclusion.

Result 5 (shifted pair lies in expanded pair). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let P be interlevel problem data. Let r and t be real numbers. Assume $r \geq 0$ (`radius`). Assume $-r \leq t$ and $t \leq r$ (`shift`). Then

$$T_t(P) \text{ is included in } X_r(P).$$

Adapted from the cited source. Scope: all bounded shifts.

Result 6 (perturbed pair is nested). [Bendich, Edelsbrunner, Morozov, and Patel, definition of perturbation pairs]
Let P be interlevel problem data. Let h be a continuous real-valued map on P.space. Assume P.interval.lower $\leq$ P.interval.upper (`order`). Then

$$A(P, h) \text{ is a nested pair.}$$

Adapted from the cited source. Scope: pair inclusion.

Result 7 (bounded perturbation lies in expanded pair). [Bendich, Edelsbrunner, Morozov, and Patel, definition of relative well groups]
Let P be interlevel problem data. Let h be a continuous real-valued map on P.space. Let r be a real number. Assume $r \geq 0$ (`radius`). Assume h is a perturbation of P bounded by r (`bounded perturbation(P, h, r)`). Then

$$A(P, h) \text{ is included in } X_r(P).$$

Adapted from the cited source. Scope: pair inclusion.

Result 8 (expanded pair has a certificate). [Topology foundations, certified expanded pair]
Let P be interlevel problem data. Let r be a real number. Assume P.interval.lower $\leq$ P.interval.upper (`order`). Assume $r \geq 0$ (`radius`). Let $T = X_r(P)$. Then

$$(\text{there exists } Q : \text{TopologicalPair}, \ Q \text{ represents } T)\,.$$

Result 9 (bounded shift has a certified inclusion). [Bendich, Edelsbrunner, Morozov, and Patel, setup for Theorem 4]
Let P be interlevel problem data. Let r and t be real numbers. Assume P.interval.lower $\leq$ P.interval.upper (`order`). Assume $r \geq 0$ (`radius`). Assume $-r \leq t$ and $t \leq r$ (`shift`). Let $S = T_t(P)$. Let $T = X_r(P)$. Then

$$(\text{there exists } Q : \text{PairInclusion}, \ Q.\text{source represents } S \text{ and } Q.\text{target represents } T)\,.$$

Adapted from the cited source. Scope: certified inclusion.

Result 10 (bounded perturbation has a certified inclusion). [Bendich, Edelsbrunner, Morozov, and Patel, definition of relative well groups]
Let P be interlevel problem data. Let h be a continuous real-valued map on P.space. Let r be a real number. Assume P.interval.lower $\leq$ P.interval.upper (`order`). Assume $r \geq 0$ (`radius`). Assume h is a perturbation of P bounded by r (`bounded perturbation(P, h, r)`). Let $S = A(P, h)$. Let $T = X_r(P)$. Then

$$(\text{there exists } Q : \text{PairInclusion}, \ Q.\text{source represents } S \text{ and } Q.\text{target represents } T)\,.$$

Adapted from the cited source. Scope: certified inclusion.

Result 11 (certification preserves pair data). [Topology foundations, certified pair constructor]
Let X be a topological space. Let S be two subsets of X. Assume S.relative $\subseteq$ S.ambient (`nesting`). Then

$$\widehat{S} \text{ represents } S.$$

Result 12 (certification preserves inclusion data). [Topology foundations, certified inclusion constructor]
Let X be a topological space. Let S and T be elements of SpacePairData. Assume S.relative $\subseteq$ S.ambient (`source`). Assume T.relative $\subseteq$ T.ambient (`target`). Assume S.ambient $\subseteq$ T.ambient (`ambient`). Assume S.relative $\subseteq$ T.relative (`relative`). Let $Q = \widehat{S \hookrightarrow T}$. Then

$$Q.\text{source represents } S \text{ and } Q.\text{target represents } T.$$

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.